

The Exact Label-Realization Radius $\nu(4, 4)$

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Abstract

For a finite integer set A , let hA be its h -fold sumset, with repetition allowed. Shin defined $\nu(h, 4)$ as the least endpoint D for which every cardinality $|hA|$ attained by a four-element integer set has a four-element representative in the literal interval $[0, D]$. His bounds give $15 \leq \nu(4, 4) \leq 16$. We enumerate, with exact integer arithmetic, all $\binom{16}{4}$ four-subsets of $[0, 15]$ and all $\binom{17}{4}$ four-subsets of $[0, 16]$. Both finite spectra are

$$\{13, 16, 17, 19, 21, 23, 24, 25, 26, 27, 29, 30, 31, 32, 33, 34, 35\}.$$

Consequently $\nu(4, 4) = 15$. The enumeration is reduced to the 35 nonnegative multiplicity vectors of weight four, and exact frequency data, source code, and a structurally different ordered-tuple replay are supplied. This resolves only the instance $h = 4$ of the label-level question.

1 Context and statement

For integers $h, k \geq 1$, define

$$hA = \{a_1 + \cdots + a_h : a_1, \dots, a_h \in A\}, \quad \mathcal{R}(h, k) = \{|hA| : A \subset \mathbb{Z}, |A| = k\}.$$

For an integer $D \geq 0$, put

$$\mathcal{R}_D(h, k) = \{|hB| : B \subseteq \{0, 1, \dots, D\}, |B| = k\}.$$

The label-realization radius considered here is

$$\nu(h, 4) = \min\{D \geq 0 : \mathcal{R}_D(h, 4) = \mathcal{R}(h, 4)\}. \quad (1)$$

Thus D is an interval endpoint, not the number of integer positions in the interval. In particular, the corresponding interval-size convention is $N(h, 4) = \nu(h, 4) + 1$.

Shin introduced (1) and proved

$$D_h^{\max} \leq \nu(h, 4) \leq \max\{h^2, D_h^{\max}\}, \quad D_h^{\max} = \binom{h+2}{2} + \mathbf{1}_{\{2|h\}} \quad (2)$$

[1, Theorem 10.3 and Corollary 10.5]. Zhang's comparison of order-preserving models and label realization records the same distinction and repeats the all- h endpoint question [2, Remark 5.2 and Question 12.5]. These two cited preprints are the closest inspected sources for the present calculation. They give the definition, the finite bounds, and the broader question; the contribution

below is the exact decision of the single instance $h = 4$. No assertion of priority or exhaustive literature coverage is made.

Substituting $h = 4$ in (2) gives

$$15 \leq \nu(4, 4) \leq 16. \quad (3)$$

Our main result chooses between these two endpoints.

Theorem 1. *Under the literal-interval definition (1),*

$$\nu(4, 4) = 15.$$

Equivalently, every cardinality of a four-fold sumset of a four-element integer set has a four-element representative in $[0, 15]$, and 15 is the least endpoint with this property.

2 Finite reduction

The spectra $\mathcal{R}_D(4, 4)$ are monotone in D and are contained in $\mathcal{R}(4, 4)$. The upper bound in (3) therefore implies

$$\mathcal{R}_{16}(4, 4) = \mathcal{R}(4, 4). \quad (4)$$

Indeed, some endpoint at most 16 realizes the full spectrum, and enlarging the interval cannot remove labels. Hence it is enough to prove

$$\mathcal{R}_{15}(4, 4) = \mathcal{R}_{16}(4, 4). \quad (5)$$

Let

$$\mathcal{C} = \{(c_0, c_1, c_2, c_3) \in \mathbb{Z}_{\geq 0}^4 : c_0 + c_1 + c_2 + c_3 = 4\}.$$

Stars and bars gives $|\mathcal{C}| = \binom{7}{3} = 35$. If $B = \{b_0 < b_1 < b_2 < b_3\}$, then

$$4B = \{c_0b_0 + c_1b_1 + c_2b_2 + c_3b_3 : (c_0, c_1, c_2, c_3) \in \mathcal{C}\}. \quad (6)$$

Both inclusions in (6) are immediate: the coefficients count the occurrences of the four elements among four summands, and every such count vector specifies a four-term sum. Thus the label of B is the number of distinct exact dot products in (6).

For $D \in \{15, 16\}$ and an integer n , define the frequency

$$F_D(n) = \#\{(b_0, b_1, b_2, b_3) : 0 \leq b_0 < b_1 < b_2 < b_3 \leq D, |4B| = n\}.$$

It follows directly that

$$\mathcal{R}_D(4, 4) = \{n : F_D(n) > 0\}. \quad (7)$$

3 Exact enumeration

Four strictly increasing integer loops enumerate each four-subset of $[0, D]$ exactly once. For each set, the calculation forms the 35 integer dot products in (6), removes duplicates, and increments the corresponding frequency. The complete output is shown in Table 1; entries not displayed have frequency zero.

The totals agree with the independently known coverage counts

$$\sum_n F_{15}(n) = 1820 = \binom{16}{4}, \quad \sum_n F_{16}(n) = 2380 = \binom{17}{4}.$$

Table 1: Exact label frequencies at the two candidate endpoints.

n	$F_{15}(n)$	$F_{16}(n)$
13	35	40
16	48	56
17	24	28
19	90	105
21	102	117
23	105	123
24	108	132
25	422	551
26	54	66
27	70	80
29	122	144
30	190	230
31	284	430
32	26	34
33	90	136
34	46	96
35	4	12
Total	1820	2380

In the machine-readable certificate, the coefficient-vector list is checked to be exactly $\mathcal{C}$; every stored witness is range-checked and has its label recomputed; and both coverage equalities above are asserted before the output is written. For example, $B = \{0, 1, 11, 15\}$ realizes the maximal displayed label 35.

As a structurally different replay, all $4^4 = 256$ ordered four-tuples in B^4 were summed for every literal four-subset B , rather than using multiplicity vectors. This reproduced both complete frequency columns. A second ancillary implementation based on nondecreasing tuples of summands also reproduced both supports. All three computations use only finite loops, sets of integers, and exact integer addition; no floating-point arithmetic or solver output enters the result.

For transparency, the following short replay is sufficient to regenerate Table 1 in any standard Python 3 implementation. The archived decisive program uses the coefficient-vector calculation above; this listing uses ordered tuples so that the two routes are visibly distinct.

```

from collections import Counter
from itertools import combinations, product
from math import comb

def label(B):
    return len({sum(x) for x in product(B, repeat=4)})

for D in (15, 16):
    freq = Counter(label(B) for B in combinations(range(D + 1), 4))
    assert sum(freq.values()) == comb(D + 1, 4)
    print(D, sorted(freq.items()))

```

4 Proof of the theorem

By Table 1 and (7),

$$\mathcal{R}_{15}(4, 4) = \mathcal{R}_{16}(4, 4) = \{13, 16, 17, 19, 21, 23, 24, 25, 26, \\ 27, 29, 30, 31, 32, 33, 34, 35\}.$$

Together with (4), this gives $\mathcal{R}_{15}(4, 4) = \mathcal{R}(4, 4)$ and hence $\nu(4, 4) \leq 15$. The lower bound in (3) gives $\nu(4, 4) \geq 15$. Therefore $\nu(4, 4) = 15$, proving Theorem 1.

5 Reproducibility, scope, and assistance disclosure

The ancillary files `coefficient_vector_recheck.py` and `coefficient_vector_recheck.json` contain the decisive exact enumeration and its full certificate. The files `trriage_enumeration.py` and `trriage_enumeration.json` contain the nondecreasing-tuple cross-check. The two locally archived arXiv HTML records identify the exact versions of the sources cited above.

The argument establishes only $\nu(4, 4) = 15$. It does not establish $\nu(h, 4) = D_h^{\max}$ for other values of h , does not concern a weak-type or order-type realization radius, and does not claim that the bounded literature inspection proves priority.

Exact computations were performed in Python. An AI system assisted with finite-search code, proof organization, cross-checking, and manuscript preparation. The mathematical conclusion is not inferred from generative output: it rests on the cited bounds, exhaustive exact-integer enumeration, archived source and output files, and a separately implemented exact replay.

References

- [1] Henry Shin. Iterated-sumset spectra: The complete exponent law and its rank geometry, 2026. arXiv:2609.01690v1 [math.CO], 1 September 2026.
- [2] Enkai Zhang. Sharp order-preserving integer models for short additive equalities, 2026. arXiv:2609.08915v1 [math.NT], 8 September 2026.